

Distributed Learning for Convex Logistic Regression

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Abstract

Our main problem statement is studying communication-efficient distributed learning algorithms for convex cost functions, with most emphasis on logistic regression (softmax). Our problem statement deals with minimizing a global convex cost function, which can be achieved by using several user equipment that sends their local weights to a central base station, which in turns does some form of averaging.

1. Introduction

The problem statement is essentially minimizing a global convex objective function of the form

$$F(w) = \frac{1}{N} \sum_{i=1}^N F_i(w) + (r(w))$$

Where

$$F_i(w) = \frac{1}{n_i} \sum_{(x_k, y_k) \in \mathcal{D}_i} \ell(w; x_k; y_k)$$

Where there are N local edge devices (from here on, abbreviated as UE, i.e, User Equipment), $\ell(\cdot)$ is a loss function, which is a function of the global aggregated weights w , and the data as well its corresponding labels. We also intend to use a regularizer (like the L^2 regularizer, for example), which still keeps it within the domain of convex cost functions.

The problem statement entails communication from the UE to a central server (from here on. abbreviated as BS, i.e, Base Station), which would help us find the global weights w by doing some operation on the local weights (usually avergaing is done [2])

1.1. Precise formulation of the problem statement

Assume there are N UE's, enumerated as $1, 2, \dots, N$. Let the entire dataset $\mathcal{D}$ be distributed across N UEs such that $\bigcup_{i=1}^N \mathcal{D}_i = \mathcal{D}$, thus, the i^{th} UE has the dataset $\mathcal{D}_i$, with n_i datapoints. Our convex problem, therefore, can be formulated as follows.

$$\min_{w \in \mathbb{R}^d} F(w) = \frac{1}{N} \sum_{i=1}^N F_i(w) + r(w), \quad (1)$$

Where

$$F_i(w) = \frac{1}{n_i} \sum_{(x_k, y_k) \in \mathcal{D}_i} \ell(w; x_k; y_k) \quad (2)$$

In the case of our interest, i.e, logistic regression, we choose to minimize the negative log of the logistic sigmoid (as we are trying to maximize the probability), i.e,

$$\ell(w, x_k, y_k) = \log(1 + \exp(-y_k w^\top x_k)) \quad (3)$$

2. The wireless communication aspect

If the wireless channel that the i^{th} UE uses has fading power gain g_i , then transmission is possible if $P_m g_i d_i^{-\alpha} \geq P_0$, where d_i is the distance of the i^{th} UE from the BS. Let us assume Rayleigh fading (thus, $g_i \sim \exp(1)$).

Thus, the probability that the i^{th} UE can participate in communication is

$$p_i = \mathbb{P}(g_i \geq \lambda d^\alpha) = \exp(-\lambda d^\alpha) \quad (4)$$

where $\lambda := \frac{P_0}{P_m}$

The transmission policy that we aim to perform is TDMA. Once the weights from the BS are sent to all UEs, we allow each UE to try and send their packet (which contains the updated local weight). In case transmission does not occur, we allow the UE to re-transmit again. This happens until either the packet is sent, or if it reaches the

maximum number of trials we allow that UE to take (if it reaches the maximum number of trials, then the packet is simply lost, and the next UE starts transmitting its weights).

To make it reasonable, we allow only a maximum of T trials altogether for all the UEs, and we can distribute these trials in some fashion between the N UEs. Say X_i is a random variable modelling the number of trials that it takes for the i^{th} UE to transmit. Let the maximum number of trials we allow the i^{th} UE to take be t_i . Hence,

$$\mathbb{P}(X_i \leq t_i) = 1 - \mathbb{P}(X_i > t_i) \quad (5)$$

$$= 1 - (1 - p_i)^{t_i} \quad (6)$$

Say we want to maximize the joint probability that all of these UEs participate (i.e, choose t_i such that the joint probability is maximum).

$$\max_{t_i} \prod_{i=1}^N \left(1 - (1 - p_i)^{t_i}\right) \quad (7)$$

Constrained by

$$\sum_{i=1}^N t_i \leq T, t_i > 0 \forall i = 1, 2, \dots, N \quad (8)$$

Since $\log(\cdot)$ is a monotonically increasing function, (7) can be written as

$$\min_{t_i} - \sum_{i=1}^N \log \left(1 - (1 - p_i)^{t_i}\right) \quad (9)$$

Thus, even for setting up the transmission, we have a convex optimisation problem at hand.

3. Solving the problem

Define $b_i := -\log(1 - p_i)$ The KKT formulation of this is

$$\mathcal{L} = \sum_{i=1}^N \log \left(1 - e^{-b_i t_i}\right) - \mu_0 \left(\sum_{i=1}^N t_i - T\right) + \sum_{i=1}^N \mu_i t_i \quad (10)$$

Since $\mu_i t_i = 0 \forall i = 1, 2, \dots, N$, it means that $\mu_i = 0$. Since we want to maximize (7), the t_i must be as large as possible, and hence, it becomes an equality (i.e, $\sum_i t_i = T$). Thus, it reduces to a Lagrangian formulation.

$$\frac{\partial \mathcal{L}}{\partial t_j} = \frac{b_j e^{-b_j t_j}}{1 - e^{-b_j t_j}} - \mu_0 = 0 \quad (11)$$

$$\implies \mu_0 = \frac{b_j}{e^{b_j t_j} - 1} \quad (12)$$

$$\implies t_j = \frac{1}{b_j} \log \left(1 + \frac{b_j}{\mu_0}\right) \quad (13)$$

From our condition, we know that $\sum_i t_i = T$, thus, we have to find μ_0 s.t

$$\sum_{j=1}^N \frac{1}{b_j} \log \left(1 + \frac{b_j}{\mu_0}\right) = T \quad (14)$$

which can be done via numerical methods. From the obtained μ_0 , we can find out t_j as well from (13)

Say w_{n+1} is the actual updated weight that would be obtained when 100% participation were to occur, and $\hat{w}_{n+1}$ be the updated weight in the actual case.

$$\hat{w}_{n+1} = \frac{1}{N} \sum_{i=1}^N 1_{\{X_i \geq t_i\}} w_{n,i} \quad (15)$$

$$\mathbb{E}[\hat{w}_{n+1}] = \frac{1}{N} \sum_{i=1}^N \left(1 - (1 - p_i)^{t_i}\right) w_{n,i} \quad (16)$$

Thus, the bias introduced to the weights is

$$Bias = |\mathbb{E}[\hat{w}_{n+1}] - w_{n+1}| \quad (17)$$

$$= \frac{1}{N} \sum_{i=1}^N (1 - p_i)^{t_i} w_{n,i} \quad (18)$$

4. Literature review

This is a summary of the literature we could find related to the topic at hand. Do note that this by no means claims to be an exhaustive list, and we hope to find more helpful papers in the future.

4.1. Federated Averaging

The paper on FedAving by McMahan et.al.[2], is a paper that deals with partitioning of datasets in an IID fashion (i.e, $\mathcal{D}_i$ are all somewhat similar. It tries to give some results in such a setting. However, the cost function at hand is non-convex, but the insights from this paper ought to be pretty useful for developing future work.

4.2. CoCoA

The CoCoA paper, by Virginia Smith et.al. [3], is more relevant to the problem at hand, as it gives convergence guarantees to convex cost functions and the framework that they have developed is also very similar to the framework developed here.

4.3. Local SGD Converges Fast and Communicates Little

This paper [4] explores the use of mini-batch SGD on the local datasets to reduce the communication overhead by relying on fewer number of communication rounds between the UE and BS.

4.4. Dual Averaging and Network Scaling

This paper by Duchi, Agarwal [1], provides more convergence guarantees and analysis based on network topologies, and this can be used for work in future potential directions.

5. Simulation and Preliminary Results

100 nodes were assumed to be distributed uniformly within a disk of radius 50m, with a path loss factor (α) equal to 2. P_m was taken to be 1mW and P_0 was taken to be -65dBm. We first solved equation (14) by using the bisection method (which is essentially binary search on a continuous set of values, as the LHS is a monotonic function of μ_0).

Also, one important point to be noted is that the number of trials is an integer, while the above problem performs continuous variable optimization. Thus, we essentially floor the value of the numbers we obtain, and we make sure that the minimum number of trials is taken as 1. (in case some of them had floor values equal to 0).

After flooring, if there is still some time budget remaining, we allot it in decreasing order of their fractional parts ($\text{frac}(x) = x - \lfloor x \rfloor$), as they were close to being rounded up anyways. In case the total time exceeds the budget (which can happen if a lot of the floored numbers, whose floor equals 0, were made into 1), then we subtract budget from those in increasing order of their fractional parts, as they deserve the extra budget the least.

The plot of accuracy v/s communication rounds that we had obtained while testing our logistic regression model for the MNIST dataset with the above set up is as follows

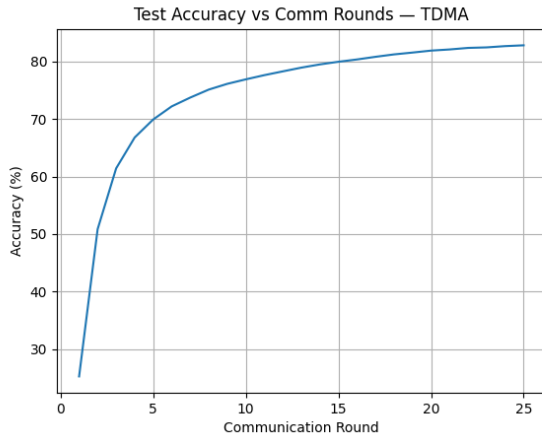


Figure 1. Accuracy over communication Rounds

References

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